

William S. Chang–Cuello

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Appointments

Greenfield Community College

2024–Pres.

Assistant Professor of Mathematics (Tenure-Track)

- Tenure-track faculty member teaching courses from College Algebra through Differential Equations, including Probability and Statistics for STEM students.
- STEM Club advisor – Mentoring students in competitive problem-solving (AMATYC) and exploring mathematical recreations and logic puzzles.
- Honors Committee advisor – Advising honors students on quantitatively modeling real-world phenomena (e.g., decomposing guitar notes into mathematical signals).

Los Alamos National Laboratory

2022–2024

Postdoctoral Researcher

- Developed and explored integral projection models to track mosquito densities across the United States as a function of weather patterns.
- Used integral modeling approaches to infer likely sources of infection from historical herd movement paths.

Rutgers University

2019–2022

Hill Assistant Professor & NSF MPS-Ascending Fellow

- Adapted mathematical and computational frameworks to ecological systems to predict long-term outcomes of communities.

Education

University of California, Davis, Ph.D. Applied Mathematics

2019

Thesis: Persistence of Single and Multispecies Systems in the Face of Environmental Uncertainty.

University of California, Berkeley, B.A. Pure Mathematics

2012

Publications

Peer-reviewed publications.

- **W.S. Cuello**, S.J. Schreiber, J.R. Gremer, P.C. Trimmer, D.L. Venable, and A. Sih. “Shifting Precipitation Regimes Influence Optimal Germination Strategies and Population Dynamics in Bet-Hedging Desert Annuals.” *The American Naturalist* 205(1): 55–75 (2025).
- R.M. Frantz, H. Godinez, K. Martinez, **W.S. Cuello**, and C. Manore. “Age-structured partial differential equations model for *Culex* mosquito abundance.” *Ecological Modelling* 494: 110764 (2024).
- **W.S. Cuello**, J.R. Gremer, P.C. Trimmer, A. Sih, and S.J. Schreiber. “Predicting evolutionarily stable strategies from functional responses of Sonoran Desert annuals to precipitation.” *Proceedings of the Royal Society B* 286(1894): 20182613 (2019).
- P.H. Crowley, P.C. Trimmer, O. Spiegel, S.M. Ehlman, **W.S. Cuello**, and A. Sih. “Predicting habitat choice after rapid environmental change.” *The American Naturalist* 193(5): 619–632 (2019).

Preprints / Submitted.

- **W.S. Cuello**, M. Gameiro, J.A. Bonachela, and K. Mischaikow. “Inferring Long-term Dynamics of Ecological Communities Using Combinatorics.” *arXiv:2309.00694* (2023). In revision for submission.

Manuscripts in preparation.

- **W.S. Cuello** and S.J. Schreiber. “Long-term Statistical Estimates of Communities in Slightly Fluctuating Environments.”
- **W.S. Cuello** and J.A. Bonachela. “Can Bet Hedging Lead Virions to Evolutionary Suicide?”
- **W.S. Cuello**, E. Romero-Severson, N. Hengartner, C. Manore, *et al.* “Predicting Disease Hot Zones from Herd Paths.”

Key Awards

- **NSF MPS-Ascending Fellowship** (DMS-2138085): Early-career federal award to develop mathematical frameworks for inferring long-term outcomes of trophic and mutualistic networks. 2021–2022
- **NJ–NExT Fellow**: Professional development program for early-career faculty sponsored by the Mathematical Association of America. 2022

Teaching Experience

Greenfield Community College

2024–Pres.

Instructor of Record: Differential Equations, Calculus I & II, College Algebra, Probability and Statistics.

Rutgers University – New Brunswick

2019–2021

Instructor of Record: Mathematical Statistics, Discrete/Probabilistic Models in Biology, Abstract Algebra I, Linear Algebra, Real Analysis I.

University of California, Davis

2012–2019

Instructor/TA: 15+ sections including Math Biology, ODEs, and Calculus.

Selected Talks & Professional Activity

- **Rocky Mountain Biological Laboratory Workshop on Integral Projection Models** (2026; upcoming): One of twelve accepted participants.
- **SIAM Annual Meeting** (2026; upcoming): Invited seminar, small-noise approximations of long-term statistical dynamics of communities of species.
- **Rutgers University** (2024): Invited seminar, using combinatorics and optimization for community dynamics.
- **Oregon State University** (2024): Invited seminar, ecological community inference.
- **Society for Vector Ecology** (2023): Invited seminar, Bayesian inference for mosquito population forecasting.
- **Montana State University** (2022): Invited seminar, determining long-term community dynamics using optimization and combinatorics.
- **Service:** Reviewer for *J. Math. Bio.*, *J. Theor. Bio.*, *Proc. Royal Soc. B.*

Technical Skills

Languages/Software: R, Python, MATLAB, C++, LaTeX, Julia.

Methods: Stochastic Modeling, Bayesian Inference, Inverse Problems, Integral Projection Models (IPMs).